



Less is more afar: responsive design for situated visualization based on viewing distance

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Abstract

Situated visualization (SitV) integrates data representation with close spatial indirection to physical referents. Viewer-referent distance affects the perceived display size for SitV, but there remains a research gap on proxemic-aware design to accommodate changing distances. To address this gap, we distill key design considerations that highlight the importance of situatedness, legibility, readability, and coherence, and conduct a quantitative experiment that reveal how the efficiency of common charts varies with viewing distance. To this end, we propose a responsive design mechanism for SitV that automatically adjusts visualization details based on viewing distances. The mechanism keeps visualizations close to referents to maintain situatedness while employing dynamic data transformation and visual mapping strategies to enhance visualization legibility and coherence. We distill the design space and develop a prototype implementation with data and visualization transformations. Two within-subject studies involving 14 and 12 participants are conducted to verify the usefulness of responsive SitV design in terms of information understanding and user engagement involving single and dual referents. The result shows that our method enables users to understand information more quickly, accurately, and with less cognitive load. Moreover, the design enhances user engagement by drawing user attention through coherent transitions of legible visualizations. Real-world applications demonstrate the usability of the prototype responsive SitV in different scenarios.

Keywords Responsive visualization · Situated visualization · Proxemic visualization · Augmented reality

1 Introduction

Augmented reality (AR) integrates digital and physical worlds, enhancing user interaction with data in their surrounding environment (Xiong et al. 2021). AR enables a semantic connection between data and physical contexts, known as situated visualization (SitV) (Kim et al. 2017). SitV presents data visualizations in close spatial indirection with the related physical referents (Moere and Hill 2012; White et al. 2006). It has gained increasing interest for its convenience and engaging experience in data exploration (Guarese et al. 2020), presentation (White 2009), and analytics (Thomas1 et al. 2018). Figure 1a illustrates SitV with bar charts overlaid on a grocery store shelf, where the shelf acts as a meaningful referent for the charts. The close proximity of the charts to the shelf enhances the situatedness experience. The design of SitV involves several key considerations, including maintaining spatial alignment with physical referents, preserving contextual relevance, and minimizing visual clutter (Willett et al. 2016; Lee et al.

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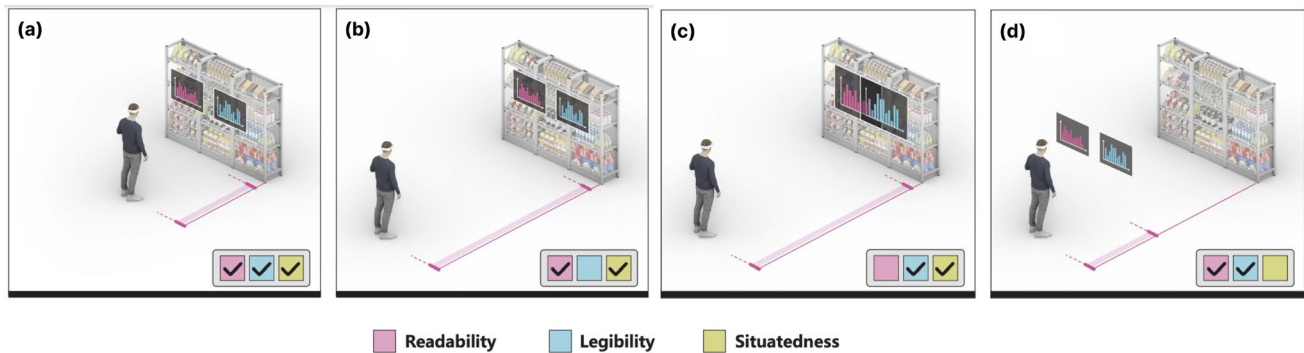


Fig. 1 Effects of viewing distance on SitV. **(a)** Close proximity maintains high readability, legibility, and situatedness. **(b)** Increased distance without any adjustment reduces legibility. **(c)** Scaling up the

charts at a distance leads to overlapping elements and decreased readability. **(d)** Adjusting the chart closer to the viewer reduces data-referent spatial relationships and thereby affects situatedness

2024). These factors are essential to support perceptual continuity and task performance in AR environments.

In a 3D environment, users have high freedom to interact with SitV by adjusting the visual angle and viewing distance. This freedom, though beneficial, introduces critical challenges for SitV design. Specifically, as users change their position, the effective visual size of data representation can shrink rapidly due to the increased viewer-referent distance. In such cases, visual elements can become unreadable or misaligned with their physical referents, thereby undermining both data comprehension and the situatedness experience. For instance, when viewing distance is changed, the increase can reduce display real estate and diminish chart legibility (Fig. 1b). Alternatively, enlarging the display size maintains legibility but may cause overlapping and hinder readability (Fig. 1c). Furthermore, relocating the charts to accommodate different viewing distances can ensure high legibility and readability, albeit at the expense of the level of situatedness, as shown in Fig. 1d.

To address these issues, we draw inspiration from responsive visualization design, which highlights the importance of automatically adapting the design and development of data visualizations to different screen sizes, devices, and orientations (Hinderman 2015; Hoffswell et al. 2020). Existing responsive visualization design mainly focuses on adapting visualizations from desktops to smaller displays such as mobile devices (Hoffswell et al. 2020; Wu et al. 2021). However, the application of responsive visualization in SitV remains largely unexplored, partly due to the added complexity of aligning visualizations with physical referents in dynamic 3D environments. Unlike traditional responsive design, which focuses on adapting to screen constraints, SitV must also account for spatial context, user perspective, and physical surroundings. These additional factors make it unclear how existing responsive techniques translate to AR-based visualizations. Therefore, there is a pressing need for design considerations for responsive SitV design. To address this need, we conduct comprehensive literature

review and factor in design considerations including *legibility* and *readability* for accurate information perception, *situatedness* for maintaining proximity to the referent, and *coherence* for consistent data representation. Particularly, *coherence* is critical for preserving a visualization's insights (Brehmer and Munzner 2013). We further conduct a quantitative experiment to evaluate the efficiency of perceiving common charts at varying viewing distances. The results indicate that detailed charts are effective at close distances, while more abstract charts are more effective at farther distances, emphasizing the importance of balancing information density and readability.

Based on these findings, we design a responsive SitV design mechanism that aligns with the design considerations and the experimental results. We apply both data and visualization transformations for four common chart types, including scatter plots, histograms, bar charts, and line charts. We then implement a prototype of these transformations created in Unity, incorporating considerations for data distribution and visual coherence. To verify the effectiveness of our design, we conduct two user studies. The first study involved 14 participants, collecting both quantitative and qualitative results for a single responsive SitV chart. While the second study, including 12 participants, tests the performance of responsive SitV for dual referents. The results demonstrate that proposed transformations enable responsive SitV charts to effectively convey information at varying viewing distances. To summarize, we make three-fold contributions:

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- We distill design considerations (i.e., legibility, readability, situatedness, and coherence) from a formative study and reveal the efficiency of common charts at varying distances in SitV.
- We design a responsive visualization mechanism with **data and visualization transformation** for common charts, implement a prototype, and demonstrate its usability through four scenarios.

- We conduct two within-subjects user studies comparing our proposed responsive SitV design and the static SitV to evaluate the effectiveness, engagement, and satisfaction of our method.

The remainder of this paper is organized as follows. Sect. 2 reviews related work on situated visualization, responsive design, and proxemic visualization. Sect. 3 presents the derived design considerations and a preliminary study. Sect. 4 describes our responsive SitV transformation strategies and implementation. Sect. 5 reports the user study. We conclude with a discussion and future directions in Sect. 6.

2 Related work

2.1 Situated visualization

Situated visualization (SitV) refers to visualizations that are presented in the physical context to which the data refers, leveraging spatial and semantic relationships between data and physical referents. The concept was first introduced by White and Feiner (2009) and further formalized by Willett et al. (2016), who defined key entities such as raw data, data representation, and physical referents. SitV is typically deployed in AR environments to enhance contextual understanding and analytic reasoning (ElSayed et al. 2016; Shin et al. 2024). SitV now has been applied across diverse fields, including tourist (Zhu-Tian et al. 2020), paper reading (Tong et al. 2023), sports analytics (Chen et al. 2023), and urban design (White and Feiner 2009). Researchers have also developed a variety of toolkits and grammars to simplify SitV creation and interaction, aiding analysis and decision-making (Cordeil et al. 2019; Büschel et al. 2021; Butcher et al. 2021; Fleck et al. 2023).

This concept of SitV differs from embedded visualization, which necessitates a close alignment between the data and its physical referent in space (Willett et al. 2016). SitV can be characterized by several critical aspects: the association between the data and its physical context, the degree to which the visualization design leverages the relevance of the data to the physical context, and the integration of the visualization's display and presentation within the physical environment (White 2009). These aspects are directly tied to the challenges of maintaining situatedness, readability, and legibility when the proximity to the referents changes. Recent work expands the concept to broader perspectives, including the dimensions of identified space, time, place, activity, and community (Bressa et al. 2022), content and ethics (Martins et al. 2023), and a more comprehensive consideration of socio-cultural, temporal, task, audience, informational, and ethical factors that shape SitV in AR, as

seen in AR decision-making examples (Martins et al. 2022). While traditional SitV assumes direct access to physical referents, proxysituated visualization (Satriadi et al. 2023) uses proxies such as digital or physical scale models. The core considerations of SitV, spatial placement, visibility, and contextual embedding, still apply, but the layered representation in proxy scenarios adds unique constraints and opportunities, requiring dedicated models.

Many prior studies in SitV have examined spatial placement and occlusion avoidance to ensure that visual elements remain visible to the user (Bressa et al. 2022), and other studies have investigated how visual design affects user attention (Morais et al. 2019) and emotional engagement (Morais et al. 2022). These factors, visibility, attentional focus, and affective response, are closely related to the legibility of visual content in context-aware environments. However, relatively little research has explored how to dynamically adapt visual properties in response to changing viewer distance or movement, a scenario common in SitV. Our work addresses the gap by introduces a responsive SitV design that adjusts data representation dynamically based on user proximity and viewing conditions, with the goal of preserving legibility and contextual relevance.

2.2 Responsive visualization

Responsive visualization involves creating dynamic visualizations that adapt to different audiences, styles, or display formats (Lee et al. 2024), particularly useful for transitioning from large screens (e.g., desktops) to smaller displays (Hoffswell et al. 2020). In traditional 2D environments, responsiveness typically focuses on adjusting layout, scale, or chart types to accommodate screen constraints while preserving the intended “message” (Lee et al. 2024). Studies examine the effects of display size, scale, and information space on comprehension and insight quality (Jakobsen and Hornbæk 2013; Badam and Elmqvist 2021). Moreover, researchers have developed tools to automatically adapt design elements such as legend size, bar width (Hoffswell et al. 2020), or adjust multi-view layouts for responsive visualization (L’Yi and Gehlenborg 2023; Zeng et al. 2024). In the case of extremely small-sized displays like smartwatches, it is more effective to adjust chart types, such as converting a bar chart into a radial layout (Blascheck et al. 2019; Brehmer et al. 2019).

However, in AR environments, “screen space” is not a fixed display area but a spatial projection that changes dynamically as users move in physical space. As users increase their viewing distance or shift angles, the effective resolution, perceived size, and visibility of visual content can vary dramatically. Unlike desktop environments, AR interfaces must adapt in real-time to spatially dependent

variables such as user proximity, orientation, and occlusion from physical objects. Prior studies have explored immersive visualization transitions by adjusting visual scale, angle, or anchoring relative to physical referents (Lee et al. 2021; Wysopal et al. 2023; Lee et al. 2023), yet these works typically focus on custom or narrative-driven designs (e.g., activity clocks Morais et al. 2019), rather than conventional visualizations such as bar charts or line graphs. These standard visualization types, which are widely used in analytic tasks, pose unique challenges in AR environments due to their reliance on precise spatial alignment and legibility across distances.

Responsive design for SitV must account for spatial relationships among the user, data representation, and physical referents. Prior studies have investigated how design decisions impact user engagement, comprehension, and reflection (Morais et al. 2019), proposing strategies such as adjusting visualization angles (Lee et al. 2021) and adapting content based on users' perspectives (Wysopal et al. 2023). Tools like Deimos (Lee et al. 2023) have been introduced to enable dynamic transitions, which are not limited to screen space adjustments but involve reconfiguring the visualization's spatial anchoring, scale, and orientation. Such transitions are inspiring in maintaining legibility as users move or change perspectives in 3D space for responsive transformations for SitV. Nevertheless, these studies mainly focus on specialized visualizations, such as the design of "activity clock" and "personal activities" (Morais et al. 2019), whilst overlooking responsive design considerations for standard visualizations which are commonly used by most users. Our contextual understanding of responsive design for SitV draws inspiration from proxemic visualization, which specifically examines how variations in viewer distance and position can inform adaptive visualization behaviors.

2.3 Proxemic visualization

Proxemic visualization, inspired by the concept of proxemics (Hall 1963), investigates how spatial factors like distance and orientation impact visual perception and interaction. While responsive visualization focuses on adapting visual representations to device or user constraints, proxemic visualization emphasizes how spatial relationships such as distance, orientation, and movement between users and visual content influence perception and interaction (Chulongsatorn et al. 2020). Early works focus on proxemic-aware interactions, such as zooming (Harrison and Dey 2008; Greenberg et al. 2011), and task mapping (Jakobsen et al. 2013). Techniques like hybrid-image visualization (Isenberg et al. 2013) and distance-based levels of detail (LoD) (Dostal et al. 2014) enable systems to dynamically adjust information density and zoom levels.

The concept of proxemic visualization has expanded to immersive environments, where proximity-based behaviors are crucial for AR interactions. In AR environments, users move freely around physical referents and virtual content. As a result, proximity such as viewing distance and user movement directly affects what information becomes visible, how it is perceived, and when it should appear, making proximity-aware design essential for maintaining situatedness and legibility. Prior studies have examined proxemic effects with virtual agents. For instance, Huang et al. (2022) observed that participants in an AR art gallery respected agents' personal space and modulated distances based on agents' gender, while Benjamin et al. (2023) found that display parameters and spatial setup influenced proxemic distances to virtual humans on transparent AR displays. These findings underscore the relevance of proxemic considerations in SitV design. Another group of studies have examined LoD techniques in dynamically adjusting visual granularity based on user proximity. For instance, label management in AR environments has been explored to enhance contextual understanding and reduce visual clutter (Köppel et al. 2021). As interface design becomes more context-aware, the need to dynamically adapt interfaces based on contextual cues becomes imperative. Notably, LoD applications in AR have demonstrated their utility in enhancing user interactions: improving compass application readability and interaction (Wysopal et al. 2023), optimizing mixed reality interfaces with cognitive load in mind (Lindlbauer et al. 2019), and creating semantically adaptive AR content in virtual reality (Qian et al. 2022).

Prior work has applied LoD to specific visualizations, such as adaptive proxemic maps (Ghaemi et al. 2022) and 3D plots (Hubenschmid et al. 2021). However, the integration of LoD with SitV for standard 2D charts operating under dynamic spatial referents remains unexplored—a gap that our research directly addresses.

3 Design consideration and preliminary study

3.1 Design considerations

Drawing from the literature, we distill four key considerations for SitV design. These considerations address the unique challenges of spatial anchoring, limited field of view, and user mobility in SitV contexts:

- **Legibility** refers to the visual clarity and perceptual distinguishability of graphical elements. To ensure legibility, charts should be designed to minimize perceptual and cognitive effort, allowing users to accurately

and completely discern information without ambiguity (Lonsdale and Lonsdale 2019). For distant viewing, keeping visualizations at their original size and format (as in Fig. 1b) may compromise legibility, as char details (e.g., bars) become less distinguishable.

- **Readability** refers to the perceived ease with which users can extract and interpret information from visual representations. Building on the framework by Cabouat et al. (2025), readability is not a singular property but a multidimensional construct shaped by the understanding of the user, the clarity of the visual layout, and the salience of data values and patterns. In this work, we focus on DataRead and DataFeat concepts, representing how easily people feel they can read data values and patterns. Visualization should be simple and display information in an accessible and easily comprehensible manner (Lonsdale and Lonsdale 2019). As users increase their distance, enlarging the display area for visualizations, as shown in Fig. 1c), may result in visualization occlusion, thereby diminishing readability.
- **Situatedness** is crucial in SitV, encompassing five dimensions (space, time, place, activity, and community) (Bressa et al. 2022). We mainly focus on space situatedness, which primarily depends on the proximity between the visualization and the physical referent, to preserve contextual relevance. A closer relationship between the visualization and its physical referent enhances the level of situatedness. When users move away, relocating visualizations closer to users but further from referents (as in Fig. 1d) can improve legibility and readability but may reduce situatedness.

These criteria have been emphasized in studies on interface design and situated visualizations in AR environments (Bressa et al. 2022; Saffo et al. 2024; Wen et al. 2023). Besides, we identify a unique consideration that pertains to the responsive visualization:

- **Coherence** in visualizations is the goal of maintaining a coherent visual style to preserve a visualization's insights despite changes in its content or data. This includes preserving datum-level information, maintaining the comparability of data points, and retaining aggregate features (Brehmer and Munzner 2013). Prior research has achieved visualization coherence through the optimization of color schemes, layout structures, and design elements (Kim et al. 2022). This uniformity facilitates users' quick association with the visualization, thereby enhancing their understanding and interaction with the data, irrespective of changes in content or distance.

Meeting all criteria simultaneously can be challenging and occasionally infeasible due to potential conflicts between them. For example, maintaining coherence often requires consistent visualization, which might impair legibility. In such cases, it may be necessary to prioritize certain design considerations over others.

3.2 Preliminary study: understanding the impact of distance

We conducted a study to investigate how viewing distance impacts the reading efficiency of charts in the context of SitV in AR environments. We aimed to establish empirical thresholds at which visualizations become inefficient, thereby informing when and how responsive designs should be applied in SitV settings. The study was designed to reflect real-world SitV scenarios (e.g., factory dashboards, museum guides, or public displays), where users often interact with visualizations anchored in physical environments and viewed from varying distances and angles. For instance, a worker may need to read sensor charts mounted on distant machinery, or a visitor might interpret data visualizations embedded in exhibits while walking through a gallery. These contexts require users to comprehend standard visualizations under dynamic spatial conditions, making distance a critical factor in legibility and interaction.

Participants. We recruited a different group of 12 participants (5 males, 7 females) aged 21–28 ($M = 23.77$, $SD = 2.21$). To ensure adequate proficiency in data visualization interpretation, we administered a brief assessment using visualizations from Mini-VLAT (Pandey and Ottley 2023). This assessment evaluated the ability of participants to read and interpret fundamental chart types. All participants demonstrated sufficient competency and had no reported visual or motor impairments. We obtained written consent from each participant prior to the study.

Study Setup. We conducted the study in a dedicated, spacious room designed for safe observation of charts from various distances without interference, as shown in Fig. 2(left). The venue has natural lighting and a simple background, minimizing visual distractions when users view the virtual charts. Before the study began, we introduced participants to the AR device and procedures, allowing them to familiarize themselves with both. The Microsoft HoloLens 2, an optical see-through HMD with a $43^\circ \times 29^\circ$ field of view and weighing 566 g, was used for the experiments. The size of the charts, set at $15\text{ cm} \times 20\text{ cm}$, was determined by referring to the HoloLens 2 manual for menu design specifications. Participants were instructed to read the charts at different distances, positioned beside the referent. We proposed a series of low-level tasks for each chart, such as *Counting the number of spattering for the scatter plot*. The

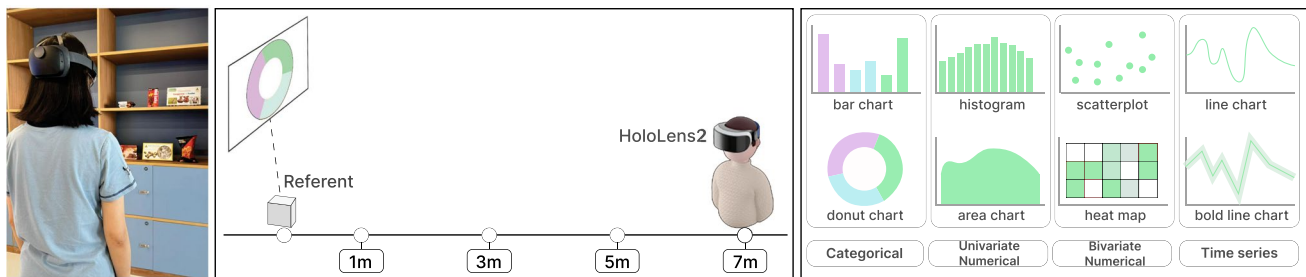


Fig. 2 Left: A user with HoloLens2 as in the preliminary study; Middle: Schematic of the study setup. The user stands at distances of 1, 3, 5, and 7 meters to observe charts and complete tasks; Right: Selected charts and their corresponding responsive designs in the study

goal was to observe performance breakdowns across distances, which can help define practical thresholds for triggering responsive transformations in situational contexts. Each participant experienced 32 trials (4 tasks $\times$ 8 charts). We recorded the time taken and the accuracy of the participants as they completed tasks by reading charts. An audio recording accompanied this process.

Determining Viewing Distances. As illustrated in Fig. 2(middle), we selected a viewing distance range of 1–7 meters according to two considerations. The first consideration was informed by the HoloLens 2 manual, which suggests that 1.25–5 meters is the recommended zone for hologram placement. We adopted this guideline to establish a realistic and representative experimental workspace, ensuring our selected distances reflect typical AR usage scenarios. The second consideration was based on a simple pilot study with 12 participants observing charts at varying distances indicating it is increasingly challenging to observe effectively beyond 7 meters. The distance is then divided into three ranges to simulate different viewing conditions: close (1 m–3 m), medium (3 m–5 m), and far distance (5 m–7 m).

Selecting Charts. We selected the following four chart types for the experiments, based on their familiarity as reported by participants in Preliminary Study 1. We adopted the responsive designs informed by previous literature, as illustrated in Fig. 2(right).

- **Bar chart.** Bar chart represents different categorical data with the height or length of each *bar* corresponding to its value. The corresponding design is chosen as donut chart, which can enhance observational efficiency when the view distance increases (Blascheck et al. 2019).
- **Histogram.** Histogram illustrating frequency distribution is generally used for univariate numerical data, using *adjacent bars*. Area chart with its area filled with color or shading was selected as the responsive design for histogram (Kim et al. 2022).
- **Line chart.** Line chart utilizes *line* mark to represent a series of data points connected by straight line segments, which is often used for observing trends over

time, identifying patterns, and visualizing fluctuations in time-series data. Thickening the line elements, rather than using a new chart type, was selected as the responsive design for line chart (Lee et al. 2024).

- **Scatterplot.** Scatter plot leverages *point* mark to represent scalar variables, which is frequently used for finding trends, outliers, and correlations in bivariate numerical data. Heat map was chosen as the responsive design for the scatterplot, following previous studies on responsive visualization (Kim et al. 2022; Wu et al. 2013).

Note that the pie chart is also regarded as a familiar chart type, but it was omitted due to the simplistic visuals and limited information, which are typically not considered for responsive design (Lee et al. 2024).

Results. Figure 3 illustrates the results for the selected charts—bar charts (left), line charts (middle), and scatter plots (right), and their responsive designs. Here, we report only the task completion time, as accuracy showed no significant differences across conditions. Overall, the responsive designs take less time to complete the tasks in all view range than the original charts. We have also identified some interesting findings.

Finding 1: Responsive design becomes necessary as viewing distance increases. Task completion times for original charts increase as the viewing distance extends from medium (5 m) to far (7 m). We attribute this performance drop to the reduced perceived size of the chart elements, which limits visual legibility. As P4 stated, "*the points and bars are too small to read. These area encodings are more visible*". This finding highlights the necessity of switching to responsive design beyond the effective viewing range where details become illegible.

Finding 2: Original charts can convey more information within a close range. Responsive designs consistently result in faster task completion times compared to original charts, influenced by task complexity and visual encoding. However, original charts are essential for detailed tasks. For instance, scatter plots often require value identification tasks, while heat maps focus on density-related tasks. As P5 stated, "*the value retrieval task is not very difficult for*

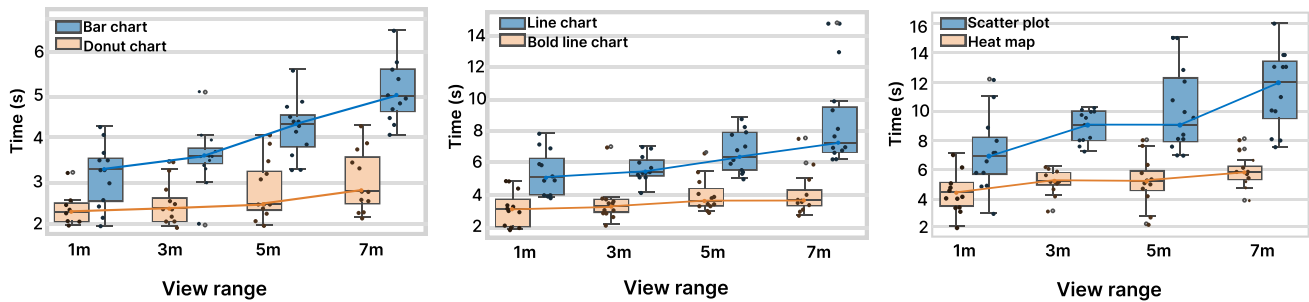


Fig. 3 Effects of view range on the efficiency of task completion for original charts and their corresponding responsive designs: (left) bar chart and donut chart, (middle) line chart and bold line chart, and (right) scatter plot and heat map

scatter plots, but it does take me some time to retrieve the values." Within the close range (1 m–3 m), the larger perceived size ensures that these fine-grained details remain resolvable. Considering the information conveyed in the original charts, most participants ($N=10/12$) suggested using the original charts within this close distance.

Finding 3: Visual encoding is the main factor. Both view range and task nature influence completion time, as seen with bar charts and scatter plots. For line charts, we only increased the line width for the responsive design, while the tasks, such as identifying trends, remained the same. The original charts encode data using positions or lines, whereas the responsive designs use area encoding. P2 noted, “The line segments are too thin; I cannot see the values. With the bold line, I can observe the trend”, highlighting the advantage of responsive adjustments.

Histogram and its responsive design exhibit similar patterns, which can be found in the supplementary material. These results help identify distance-dependent performance drops for the charts, providing empirical evidence for when responsive visualizations offer measurable benefits.

4 Responsive design for situated visualization

This section outlines the responsive visualization mechanism (Sect. 4.1) followed by prototype implementation (Sect. 4.2).

4.1 Responsive design mechanism

The findings from Sect. 3.2 indicate minimal performance differences between original and responsive charts at close distances. However, as viewing distance increases to medium range, the impact on reading performance becomes more pronounced. At far distances, chart reading efficiency improves significantly with responsive designs. Beyond the comfort zone, a cognitive chart takes precedence over information, prompting yet another transformation to the

responsive charts. We define two transformations at medium and far distances respectively.

- Data Transformation.** Our preliminary study indicates that chart reading performance declines notably as the viewing distance increases. This degradation is primarily due to the decreased perceived size, where retaining the original data density results in severe visual clutter and reduced legibility. To address this, we prioritize data transformation strategies that adapt information granularity and maintain coherence across chart types. Specifically, two methods are used: sampling, which reduces data volume by selecting representative points to preserve distribution clarity, and aggregation, which merges data points to maintain overall trends while minimizing clutter.
- Visualization Transformation.** When the viewing distance exceeds 5 m, the chart reading performance would be notably affected by distance ranging. Viewers primarily seek a comprehensive overview of related data within 5 m–7 m. Responsive designs help reduce cognitive load at greater distances by transforming visualizations to adapt chart types and scale visual elements. Donut charts, area charts, and heatmaps offer better legibility than bar charts, histograms, and scatter plots due to their simplified, aggregated presentations. Moreover, during such transformations, maintaining a coherent transition is essential to avoid loss of information for viewers. To ensure consistent visual encodings, another strategy is to scale up the visual element (such as increasing line weight) without causing occlusion.

Responsive design mechanisms for the common charts are following as illustrated in Fig. 4.

- Bar Chart.** The transformations for bar charts are illustrated in Fig. 4a. The first transformation, data aggregation (Kim et al. 2022), reduces the number of bars and simplifies visual complexity. The second transformation turns the bar chart into a donut chart (Blascheck et al.

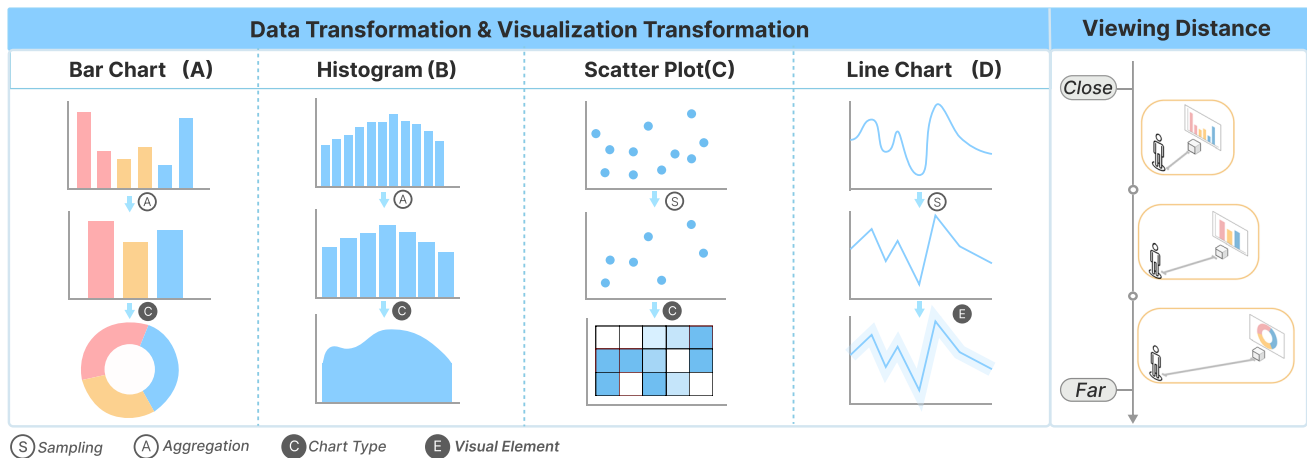


Fig. 4 Responsive design mechanism for SitV. We derive design patterns for data and visualization transformations of four common charts: (a) bar chart; (b) histogram; (c) scatter plot; and (d) line chart

2019), which shifts the focus toward categorical distribution. This transformation is motivated by the robustness of donut charts under increased viewing distance—offering a more stable and glanceable encoding when both perceptual load and visual angle become limiting factors.

- Histogram.** As depicted in Fig. 4b, we propose the data transformation stage for histograms involves aggregation, reducing bar numbers (Kim et al. 2022), and segmenting data. The visualization transformation stage changes the histogram into an area chart, focusing on the overall trend. The area chart, with its simpler encoding, effectively communicates key patterns and trends from a distance.
- Scatter Plot.** As depicted in Fig. 4c, through data transformation, specifically sampling (Bertini and Santucci 2006), we reduce the number of data points, enabling a more distinct representation of data distribution. We then transform the scatter plot into a heatmap (Kim et al. 2022; Wu et al. 2013) through visualization transformation. This further reduced detail, while focusing on data distribution. The heatmap, with the simpler encoding, becomes more legible from a distance.
- Line Chart.** We design a transformation for line charts used for time series data, as shown in Fig. 4d. The data transformation, specifically sampling (Setlur and Chung 2021), reduces the number of data points and smoothens the line. The visualization transformation involved gradually thickening (Lee et al. 2024) the line to ensure readability from a distance.

4.2 Prototype implementation

We developed AR prototypes to demonstrate responsive chart designs that adapt automatically to user distance

from the SitV. Built for HoloLens 2 using C# in Unity 2022.3.14f1c1 LTS with MRTK-Unity, the prototypes implement transformation patterns and detect distance changes to trigger transitions at predefined points. The implementation leverages HoloLens 2 specifications, including 2k resolution, holographic density >2.5k, and a 60 FPS target frame rate, ensuring performance and clarity. The distance thresholds for triggering data and visualization transformations were decided by participants according to their personality and different definitions of distance thresholds. We utilize the built-in spatial tracking capabilities of HoloLens 2 to continuously measure the distance between the user's head position and the visualization object. Transformations are automatically triggered when the user's distance enters predefined threshold ranges. These thresholds do not adapt to environmental factors such as indoor or outdoor settings. To ensure robustness in real-world AR environments, we tested the prototypes under diverse lighting conditions, including indoor low-light, indoor strong light, and natural daylight environments. These prototypes effectively showcase how responsive visualizations dynamically adjust to enhance data accessibility across varying viewing distances and under different lighting conditions.

Figure 5 illustrates the scatter plot transformation mechanism we implemented in Unity. Raw data in CSV format was processed using C# scripts to create scatter plots while documenting the position of each data point. Sampling is performed dynamically during visualization, adjusting to changes in the dataset in real time. The process begins with computing grid density to quantify the number of data points within each grid cell. Next, the target sample count was calculated to determine the number of sample points needed, guided by the predefined sampling rate. The final step of this phase involved deciding whether to retain or eliminate data points from the grid based on their density, while ensuring

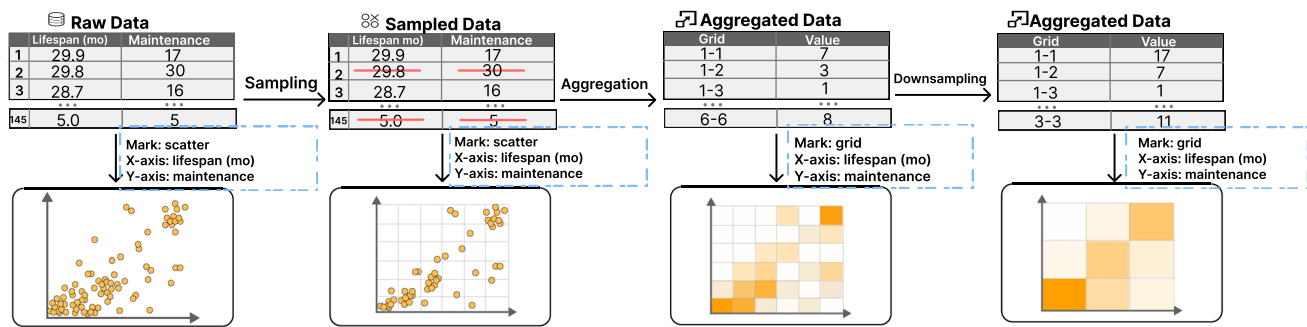


Fig. 5 Prototype implementation for scatter plot. During the data transformation phase, the raw data is first sampled to reduce the scatter point quantity. During the visualization transformation phase, scatter

points are aggregated to facilitate transformation into a heat map. As the viewing distance extends, the heat map's resolution is progressively decreased via downsampling

that the remaining scatter point information was preserved. In the aggregation phase, the remaining points were mapped onto a 2D grid. A density estimation process assigned heat-map colors, which were interpolated and projected onto a 2D texture to form the heatmap. During the downsampling phase, the resolution of the heat map was modulated following distance variations. The heat map texture and colors were updated in line with newly sampled data.

For other chart prototypes, we implemented similarly: 1) For the bar chart, we initially filtered the raw data, selecting specific hierarchical data from the child nodes of the root node based on distance, and subsequently generated a bar chart. When the distance exceeded a user-defined threshold, the bar chart data was archived to generate a donut chart; 2) For the histogram, after inputting the raw data set and generating the corresponding histogram, we recalculated the data for each bar post the first transformation and regenerated the histogram. Following the second transformation, we used the aggregated data from the histogram to redraw the area chart; 3) For the line chart, we employed the Ramer-Douglas-Peucker algorithm during the sampling process to simplify the line. The thickness of the line was then controlled by mapping the distance value. Although the system implementation relies on established visualization techniques, its primary contribution lies in demonstrating how these transformations can be embedded into a responsive AR visualization pipeline. Our goal was to validate the feasibility of responsive SitVs through functional prototypes rather than to introduce novel interaction or rendering algorithms.

5 Evaluation

5.1 Case study

All four scenarios were implemented using the same underlying prototype without requiring any scenario-specific

modifications. This demonstrates the generalizability of our responsive mechanism across diverse real-world tasks.

Scenario 1: Book Borrowing. To assist readers in navigating a large library collection, a cogent graphical representation is important. In this case, we employed bar charts and the corresponding design space to represent book counts by classification on two bookshelves. As depicted in Fig. 6a, the granularity of the book categories is adapted to viewing distances. At close range, detailed book categories are displayed for precise exploration. As the distance increases, the bar chart aggregates into broader categories, providing an overview of the collection. From afar, a corresponding donut chart will be presented to the user, providing an overview of the book categories with simplified visualization. This scenario required the integration of hierarchical classification data from the library system, which led us to adjust the bar chart structure to support on-demand aggregation by classification levels.

Scenario 2: Goods Sales Monitor. Monitoring supermarket sales often requires detailed analysis, which can be improved through effective visualizations. Histograms are particularly suitable for this purpose, as they display the relationship between sales volumes and dates on two shelves in a supermarket. As shown in Fig. 6b, the x-axis represents thirty dates within a month and the y-axis shows the sales of the goods. Each bin corresponds to daily sales. At close range, the histogram shows detailed daily sales with thirty bins. At medium range, the bins are grouped to represent three-day intervals, offering a clearer monthly overview. From afar, an area chart replaces the histogram, providing a simplified summary of sales trends for quick insights. This scenario introduced challenges in balancing temporal resolution and readability. We refined the binning algorithm to dynamically adjust bin widths at intermediate distances, ensuring consistent visual density across transitions.

Scenario 3: Equipment Management. A comprehensive record of equipment usage and maintenance is vital for efficient management and student borrowing. Fig. 6c



Fig. 6 We showcase four usage scenarios to demonstrate the usability of our responsive visualization mechanism in real-world scenarios: **(a)** book borrowing; **(b)** goods sales monitor; **(c)** equipment management; and **(d)** electricity monitor

demonstrates a responsive scatter plot design, showing equipment durability and maintenance needs on a side of equipment shelf. The x-axis represents usage in months, while the y-axis indicates maintenance frequency. At close range, managers can view detailed data for each piece of equipment, with each point representing an item. At a middle distance, the scatter plot highlights data distribution, trends, and clusters. From afar, it transforms into a heatmap, summarizing maintenance patterns. Unlike previous scenarios, this dataset was highly sparse and non-uniform. To address this, we adjusted the heatmap color mapping and grid resolution to better reveal clusters and outliers during aggregation.

Scenario 4: Electricity Monitor. Efficiently managing dormitory buildings requires monitoring and analyzing room electricity consumption. A suitable data visualization, such as a line chart, is essential for this case. A line chart is an effective visualization tool for this purpose, displaying power consumption (y-axis) over time (x-axis) while accommodating the complexity of floors and areas. As shown in Fig. 6d, the original line charts will be shown at the close viewing distances with detailed information on four different parts of dormitory. As the viewing distance increases, the original line will be sampled to show the main trends of consumption in a period. From afar, the sampled lines are thickened to reduce cognitive load. This responsive

SitV enables to increase in efficiency, making it a practical solution for dormitory electricity monitoring. This scenario involved multiple overlapping lines from different rooms, which led us to customize the line thickness mapping and adjust the sampling granularity to reduce clutter at long-range views.

5.2 User study 1: single referent

We conducted a user study to investigate the effectiveness of the responsive SitV with single referent. Through task completion and questionnaire surveys, we obtained both quantitative and qualitative results for further analysis.

Participants. We recruited 14 participants (P1-P14) (7 males and 7 females; aged between 22 and 38, with an average age of 25.3) for our study. All participants were university students and staff members from varying backgrounds, including data science, art, material science, physics, bioinformatics, and psychology. This participant pool was different from the preliminary study, and none had taken part in the previous study. Two participants were familiar with AR techniques, and six had AR experience. However, only three had experience with HoloLens 2. None of the participants had any sight or mobility impairments. For each scenario, we guided the participants in observing the SitV from various distances and completing corresponding tasks.

The study was approved by the authors' institution's ethics board, with all participants providing informed consent.

Experimental Conditions. In our design, each chart undergoes two transformations based on viewing distance changes. We set up four different conditions in our user study to compare the performance of each individual responsive design and the complete design, as follows:

- Static acts as the baseline control group, with *no transformation* of the visualization based on the user's viewing distance, remaining in its original state.
- In DataOnly, the chart undergoes *only data transformation* at medium distances, without any visualization transformation. This is conducted to investigate the efficacy of data transformation and to compare the impacts of standalone data transformation against combined data and visualization transformation.
- VisualOnly involves *only visualization transformation*, without any data transformation. This allows us to evaluate the individual effectiveness of visualization transformation at extended distances and compare it with the combined effect of data and visualization transformation.
- In Combined, the SitV chart experiences *both data and visualization transformations*, enabling users to

perceive the sequential alterations that adapt to their viewing distance.

To ensure the rigor of the user study and prevent inaccuracies in the experimental results due to the order of the conditions, the conditions were counterbalanced across participants.

Procedure and Tasks. Participants were first briefed on the experiment and provided informed consent. They then completed a pre-study survey assessing their demographic background, prior experience with AR, and familiarity with the four chart types, including bar charts, histograms, scatter plots, and line charts. Eleven participants indicated high familiarity with these chart types, while three reported moderate familiarity. They were then introduced to the experimental design and guided through tasks using HoloLens 2. The task set was designed based on a synthesis of prior research on data visualization in immersive environments. These tasks reflect typical user interactions with SitV in real-world scenarios (e.g., equipment management, electricity monitoring), each mapped to realistic application contexts. To address individual differences in judging close, medium, and far viewing distances for SitV visualization, and the degree of information participants wished to obtain at different distances, a preliminary test was conducted to determine individual transition points before the user study. These points, averaging 3.26m (data transformation) and 5.32m (visualization transformation), were kept consistent during the study to avoid bias. They are aligned with our settings for medium and far distances, refer to preliminary study (Sect. 3.2). During the test, participants viewed SitV visualizations and moved in a straight line. When a participant first requested a visualization transformation, the distance recorded by the HoloLens 2 camera was set as the first transition point, triggering a data transformation. A second request during further movement established the second transition point, initiating a visualization transformation.

We counterbalanced the order of conditions across participants, testing all combinations of experiment conditions (4 scenarios $\times$ 4 conditions $\times$ 6 tasks), with each participant completing 96 trials across four scenarios. In each scenario, the displayed dataset was directly related to the surrounding environment, though explicit text labels were hidden during tasks to minimize semantic priming. Participants began by observing the scatter plot in the SitV context within a simulated device management room and performed tasks under four conditions. In each experimental process, we posed six questions to the participants. Each scenario included six visualization-based questions, such as comparing, identifying trends, or finding extremes. For example, *Which is more, red books or green books?*, refer to Fig. 6a.

Questions focused on the main characteristics of visualization, avoiding tasks involving identifying numbers or labels. A complete list of the questions for each visualization type is provided in the supplementary to support full replication of the study. Participants were encouraged to move freely to choose optimal viewing distances and provide concise answers. After each scenario, they completed a questionnaire and provided feedback, with rest breaks between scenarios to ensure validity. Each session lasted 1.5 h, with participants compensated \$10, and all sessions were recorded.

Measures. We evaluated the performance of the four conditions by analyzing the accuracy and efficiency of the participants in completing the tasks and the results of the questionnaire scores in the experimental records. Efficiency was measured by the time it took for participants from the end of the task introduction to the start of answering. Accuracy was measured by the correctness rate of the answers by analyzing the audio recordings. Qualitative user feedback, including satisfaction, engagement, usefulness, and usability of the design space, was also collected. Viewing distance data was automatically captured and recorded by the HoloLens 2 camera.

Hypotheses. We base our expectations on the principle that both data and visual transformations are needed to preserve information salience and perceptual accessibility at different distances. Our proposed method is also supposed

to generate data and visualization transformations that meet the design requirements. According to the above, we hypothesize that:

- **H1.** For all charts and tasks, Combined is hypothesized to be more efficient and achieve higher accuracy than Static, DataOnly, and VisualOnly.
- **H2.** For all charts and tasks, we hypothesize that DataOnly and VisualOnly would be more efficient and achieve higher accuracy than Static.

Results. We evaluated the effectiveness of the design space for each chart across the four conditions by analyzing task accuracy (AbsDiff) and completion time (seconds) with 95 percent confidence. Figure 7 illustrates AbsDiff (top) and completion time (bottom) for each condition across the charts. Due to non-normal data distribution, we applied a Friedman test to assess the significant effects of conditions on task accuracy and time at a significance level of $\alpha = 0.05$. Furthermore, we ran a post-hoc Wilcoxon signed-rank test to perform pairwise comparisons of different conditions. To control for Type I error due to multiple comparisons, we applied the Bonferroni correction to adjust the significance threshold. Significant values are reported for $p < .05(*)$, $p < .01(**)$, $p < .001(***)$.

Hypothesis 1. H1 is partially confirmed by evaluating the overall accuracy (AbsDiff) and completion time

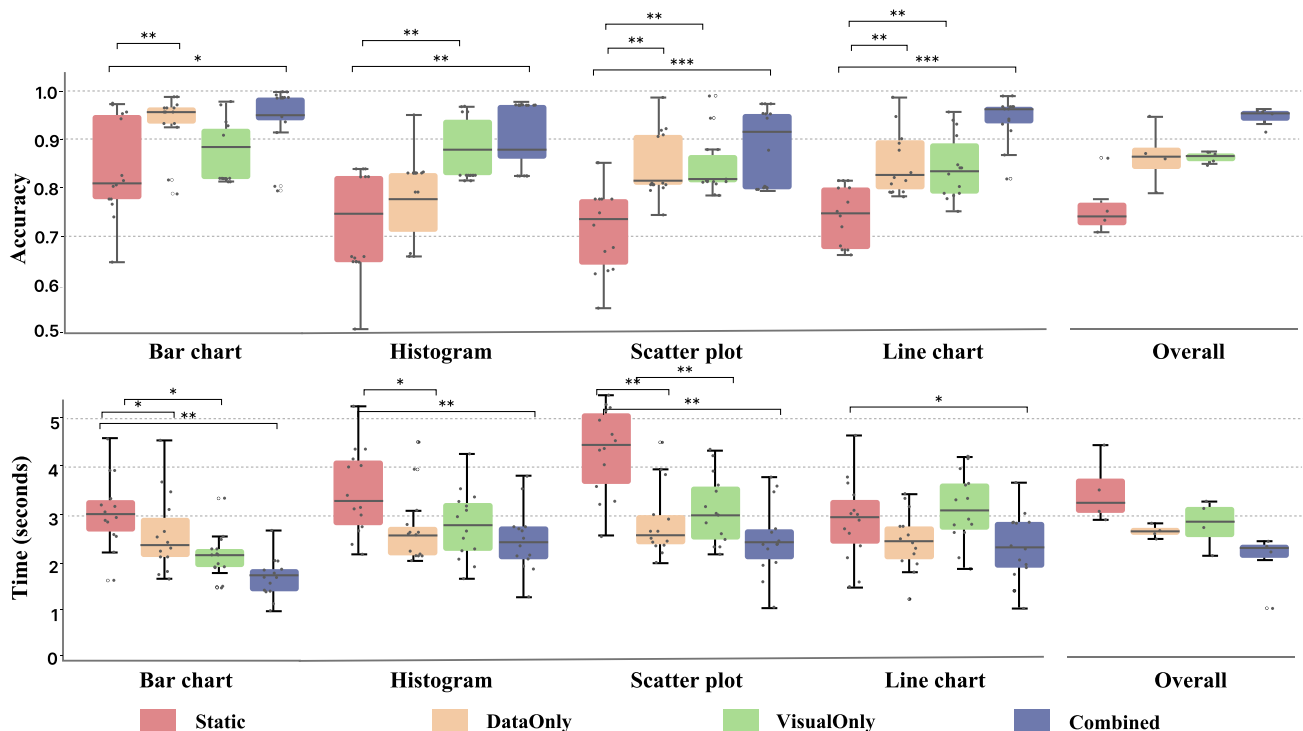


Fig. 7 Average accuracy (top) and completion time (bottom) with 95% confidence intervals for four conditions: static group, data transformation group, visualization transformation group, and the whole transformation group, with single referent

across the four conditions for each chart. A Friedman test revealed significant effects of conditions on accuracy for all chart types ($\chi^2(3) = 16.55, p = 0.00087$; $\chi^2(3) = 29.44, p = 1.807e - 06$; $\chi^2(3) = 23.14, p = 3.782e - 05$; $\chi^2(3) = 25.84, p = 1.033e - 05$ (***)). Post-hoc Wilcoxon signed-rank tests showed that Combined outperformed Static ($Z = -2.51, p = 0.011$) for bar charts (*) and significantly outperformed Static ($Z = -3.3, p = 0.0024$) for histograms (**). For the scatter plot and line chart, Combined significantly outperforms Static ($Z = -3.23, p = 0.00014, Z = -3.3, p = 0.0011$) (***). In terms of completion time, significant differences were found for bar charts ($\chi^2(3) = 30.086, p = 1.324e - 06$), histograms ($\chi^2(3) = 22.714, p = 4.632e - 05$), scatter plots ($\chi^2(3) = 31.2, p = 77.715e - 07$), and line charts ($\chi^2(3) = 17.24$). After a further post- $p = 0.00063$ hoc Wilcoxon signed-rank test. Regarding the bar chart, Combined significantly trimmed the completion time compared to Static ($Z = 3.23, p = 0.0014$) (**) and DataOnly ($Z = 2.54, p = 0.015$) (*), while slightly to VisualOnly ($Z = 3.3, p = 0.0011$) (**). In the case of the histogram, the completion time under Combined is notably less than that under Static ($Z = 3.17, p = 0.0017$) (**). For the scatter plot, Combined dramatically outperforms Static ($Z = 3.3, p = 0.0011$) (**), illustrating that the completion time was significantly reduced under Combined. However, only a little differences in completion time were observed for line charts between the Combined and the Static conditions ($Z = 2.23, p = 0.028$) (*). Overall, Combined achieved better accuracy and faster completion times for most charts, partially confirming H1.

Hypothesis 2. The confirmation of H2 is achieved by analyzing the average overall accuracy and completion times across three conditions for each chart. Using a similar method, we find the influence of conditions on accuracy between DataOnly and VisualOnly in comparison to Static. For the bar chart, DataOnly slightly outperforms Static ($Z = -3.3, p = 0.0016$) (**). In the case of the histogram, only VisualOnly outperforms Static ($Z = -3.3, p = 0.0024$) (**). Both DataOnly ($Z = -3.3, p = 0.0017$) and VisualOnly ($Z = -3.3, p = 0.0016$) (**) for the scatter plot are significantly more accurate than Static. For the line chart, both DataOnly ($Z = -3.08, p = 0.0028$) (**) and VisualOnly ($Z = -3.11, p = 0.0021$) (**) are also more accurate than Static. Regarding task completion time, VisualOnly outperformed Static for bar charts ($Z = 3.23, p = 0.012$) (*), DataOnly was also slightly faster than Static ($Z = 2.54, p = 0.012$) (*). For histograms, DataOnly outperformed Static ($Z = 2.29, p = 0.035$) (*). Both DataOnly ($Z = 3.3, p = 0.0013$) and VisualOnly ($Z = 3.23, p = 0.0014$) showed better completion times compared to Static for scatter plots (**). For the line chart, both DataOnly and VisualOnly show no significantly better

performance than Static. In summary, DataOnly and VisualOnly for most charts achieve better performance than Static, partially confirming H2.

5.3 User study 2 : dual referents

The experiment with single-chart tasks primarily examines efficiency and accuracy of distance-responsive visualizations. We further designed a new study to better capture the situatedness of SitV in context. In this extended study, participants interacted with two co-located visualizations, each associated with distinct referents, and performed comparative and referent-identification tasks.

Participants. We recruited 12 new participants (P15-P26) (8 males and 4 females; aged between 21 and 29, with an average age of 24.6), with similar academic backgrounds as in the first study. None of them had participated in the previous experiment. Three participants reported familiarity with AR techniques, five had prior AR experience, and two had used HoloLens 2 before. All participants had normal or corrected-to-normal vision and no mobility impairments.

Procedure and Tasks. In the extended study, the procedure remained the same, but participants interacted with two co-located situated visualizations, each associated with distinct referents (e.g., two bookshelves or two devices). The tasks were accordingly expanded to include referent recognition and comparative reasoning between visualizations. For example, participants were asked to “*Compare the two bookshelves — which has a higher total quantity of red books?*”

- **H3.** In the study involving two referents, we hypothesize that Combined will remain the most efficient condition and achieve the highest accuracy across all charts and tasks.
- **H4.** In the study involving two referents, we hypothesize that DataOnly and VisualOnly would be more efficient and achieve higher accuracy than Static.

Results. **Hypothesis 3.** Across all visualization types, statistical analyses confirmed H3 for both efficiency and accuracy as 8 showing. For the *bar chart*, efficiency differed significantly across conditions ($\chi^2(3) = 18.795, p = 0.0003$) (***), with post-hoc tests showing that participants were more efficient in the Combined mode ($Z = 2.12, p = 0.003$) (**). Accuracy also varied significantly ($\chi^2(3) = 19.402, p = 0.00023$) (***), where the Combined mode yielded the highest performance ($Z = 2.51, p = 0.011$) (*). In the *histogram* task, efficiency showed a reliable difference ($\chi^2(3) = 10.7, p = 0.013$) (*), and participants were again faster in the Combined condition ($Z = 2.98, p = 0.0026$) (**). Accuracy exhibited a robust

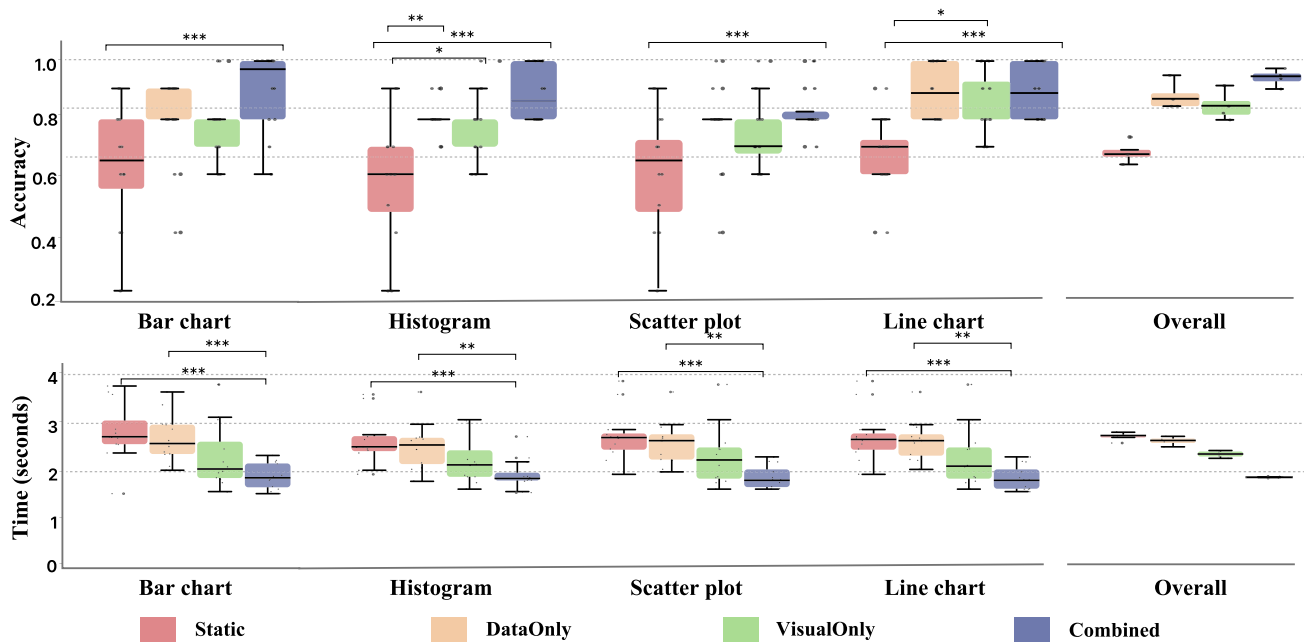


Fig. 8 Experiment results for responsive design with dual referents

effect ($\chi^2(3) = 20.969, p = 0.0001$ ***), with the Combined mode outperforming all others ($p = 0.00095$ ***). For the *scatter plot*, efficiency differences were highly significant ($\chi^2(3) = 24, p = 2.5 \times 10^{-5}$ ***), and the Combined mode led to faster task completion ($Z = 3.06, p = 0.00074$ ***). Accuracy also varied across conditions ($\chi^2(3) = 18.489, p = 0.00035$ ***), with the Combined mode significantly surpassing Static and DataOnly ($p = 0.0032$ **). Finally, the *line chart* revealed similar patterns. Efficiency was significantly affected by condition ($\chi^2(3) = 25.3, p = 1.34 \times 10^{-5}$ ***), with the Combined mode providing the best performance ($p = 0.00074$ ***). Accuracy also improved under the Combined setting ($\chi^2(3) = 14.777, p = 0.0020$ **), outperforming the Static condition ($p = 0.014$ *). Overall, these results consistently demonstrate that Combined mode enhances both efficiency and accuracy across all chart types.

Hypothesis 4. According to the results by post-hoc pairwise Wilcoxon tests (Bonferroni-corrected), H4 is partially confirmed. For *bar chart*, the result indicated no significant difference between *Static* and *DataOnly* or between *Static* and *VisualOnly*. These results partially support H4, showing that while *DataOnly* and *VisualOnly* did not significantly outperform *Static*, participants achieved notably better performance in the *Combined* mode. The results of the *histogram* showed that both *DataOnly* ($Z = 2.98, p = 0.0026$ **) and *VisualOnly* ($Z = 2.52, p = 0.0369$ *) significantly outperformed the *Static* condition in accuracy. Similarly, for *scatter plots*, the results indicated that both *DataOnly* ($Z = 3.1, p = 0.0032$ **) and *VisualOnly* ($Z = 2.02, p = 0.0169$ *) were sig-

nificantly more accurate than the *Static* condition. The result of *line chart* showed no significant differences between *Static* and *DataOnly* ($Z = 0.71, p = 0.134$) or *Static* and *VisualOnly* ($Z = 1.88, p = 0.065$ *)). However, participants achieved significantly higher accuracy in the *Combined* condition compared to *Static* ($p = 0.014$ *)). These findings partially support H4, indicating that while *DataOnly* and *VisualOnly* did not differ significantly from *Static*, the integration of both data and visual cues in the *Combined* mode led to improved accuracy.

5.4 User feedback

We collected user feedback on Combined, gathered via questionnaires and short semi-structured interviews. The questionnaire was developed based on established instruments in visualization and HCI research, drawing on constructs for utility, usability, workload, engagement, and satisfaction. Each construct was assessed with 1–2 items rated on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree), and all items were positively worded to reduce cognitive bias. Participants completed the questionnaire immediately after tasks for each chart type across all four experimental conditions (Static, DataOnly, VisualOnly, and Combined). While this section primarily reports the qualitative results for the Combined condition, the full comparative ratings for all conditions are provided in the supplementary material. In addition, we conducted brief follow-up interviews (2–4 min per participant) to collect qualitative feedback. These semi-structured interviews invited

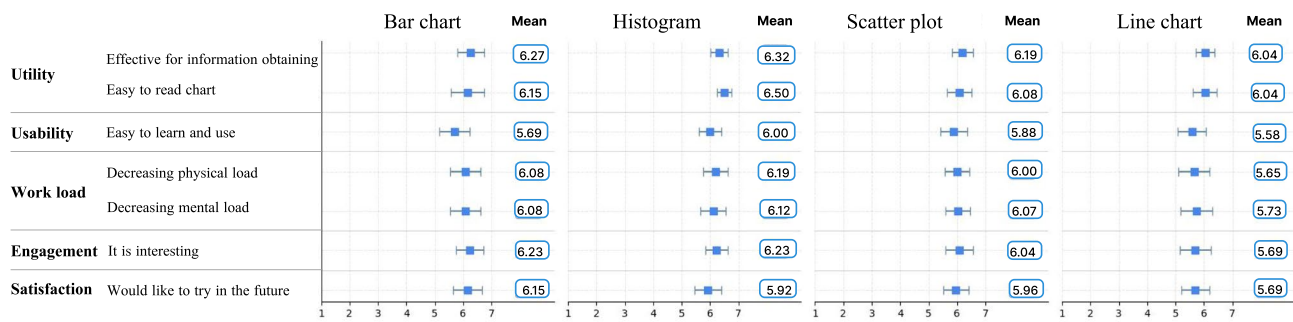


Fig. 9 Qualitative results on utility, usability, workload, engagement, and satisfaction of our responsive SitV design mechanism

participants to elaborate on their ratings, describe positive and negative experiences, and suggest improvements (e.g., interface layout, visual transformation timing). Detailed ratings are presented in Fig. 9. Combined generally scored higher than 5 on average, indicating the users' satisfaction with our responsive designs.

- Utility.** Participants generally agreed that the responsive SitV designs were effective and helpful in supporting information acquisition and chart readability across all visualization types. The bar charts and histograms were rated as particularly effective for obtaining information ($\mu = 6.27$, 95% CI = [5.80, 6.73]; $\mu = 6.32$, 95% CI = [6.01, 6.63]) and easy to read ($\mu = 6.15$, 95% CI = [5.57, 6.73]; $\mu = 6.50$, 95% CI = [6.24, 6.76]). Scatter plots also showed strong utility ratings ($\mu = 6.19$, 95% CI = [5.83, 6.55]; $\mu = 6.08$, 95% CI = [5.64, 6.52]), indicating that participants found them useful and easy to interpret. Line charts received slightly lower yet still positive evaluations ($\mu = 6.04$, 95% CI = [5.71, 6.37]; $\mu = 6.04$, 95% CI = [5.62, 6.46]), suggesting consistent usability across chart types. Participants attributed the high utility to the “responsive transformations that maintain clarity at varying distances” (P4), and P17 mentioned that these designs “make it easier to focus on data without worrying about visibility.”
- Usability.** Participants agreed that the responsive SitV designs were easy to learn and use across all visualization types. Bar charts ($\mu = 5.69$, 95% CI = [5.16, 6.23]) and histograms ($\mu = 6.00$, 95% CI = [5.60, 6.40]) received the highest usability ratings, indicating that participants quickly adapted to the responsive transformations. Scatter plots also showed favorable usability ($\mu = 5.88$, 95% CI = [5.41, 6.36]), while line charts received slightly lower but still positive evaluations ($\mu = 5.58$, 95% CI = [5.08, 6.08]). Overall, participants found the responsive interactions intuitive, noting that the designs “automatically adjust to distance changes without additional operations” (P20) and are “straight-forward to understand after a short time of use” (P2).
- Workload.** Participants reported that responsive SitV designs effectively reduced both physical and mental load during visualization tasks. Bar charts ($\mu = 6.08$, 95% CI = [5.54, 6.61]) and histograms ($\mu = 6.19$, 95% CI = [5.75, 6.64]) were perceived as most effective in decreasing physical strain, while scatter plots ($\mu = 6.00$, 95% CI = [5.57, 6.43]) and line charts ($\mu = 5.65$, 95% CI = [5.11, 6.20]) showed slightly lower but still positive evaluations. A similar pattern was observed for mental load, with histograms ($\mu = 6.12$, 95% CI = [5.67, 6.56]) and bar charts ($\mu = 6.08$, 95% CI = [5.54, 6.61]) rated highest, followed by scatter plots ($\mu = 6.04$, 95% CI = [5.60, 6.47]) and line charts ($\mu = 5.73$, 95% CI = [5.16, 6.30]). Participants noted that while some physical effort was required to move closer to the charts, the responsive transformations helped minimize unnecessary movement and visual strain. One participant mentioned experiencing occasional “physical exhaustion due to the headset weight” (P7), but most agreed that the overall workload was manageable and improved by the adaptive design.
- Engagement.** Participants expressed strong interest and engagement when interacting with the responsive SitV designs. Histograms ($\mu = 6.19$, 95% CI = [5.80, 6.58]) and bar charts ($\mu = 6.08$, 95% CI = [5.63, 6.54]) were rated as the most engaging, followed closely by scatter plots ($\mu = 6.00$, 95% CI = [5.57, 6.43]) and line charts ($\mu = 5.77$, 95% CI = [5.29, 6.26]). Participants reported that the adaptive adjustments enriched their experience, making the visualizations “more dynamic and enjoyable to use” (P9) and “interesting to explore from different distances” (P15). Several participants also mentioned that the responsive behavior increased their curiosity and attentiveness, encouraging them to interact with the charts more actively.
- Satisfaction.** Overall, participants showed high satisfaction with the responsive SitV designs and expressed interest in using them again in the future. Histograms ($\mu = 6.27$, 95% CI = [5.89, 6.65]) and bar charts ($\mu = 6.12$, 95% CI = [5.69, 6.55]) received the

highest ratings, indicating strong willingness to continue exploring these responsive visualizations. Scatter plots ($\mu = 6.04$, 95% CI = [5.60, 6.47]) and line charts ($\mu = 5.85$, 95% CI = [5.33, 6.37]) also demonstrated positive responses, suggesting broad acceptance across all chart types. Participants appreciated how the designs enhanced comfort and interaction, with comments such as “I would like to use this kind of responsive chart again in future projects” (P23). These results indicate strong satisfaction and sustained interest in adopting responsive visualizations for future use.

6 Discussion

The study demonstrates that combining data and visualization transformation (Combined) outperforms applying data transformation (DataOnly) or visualization transformation (VisualOnly) alone, supporting prior research on responsive SitV. Data transformation is most effective at medium distances, balancing mental load and coherence, while visualization transformation excels at far distances, beyond the comfort zone. Together, these methods improve chart readability and legibility, offering a more efficient approach to data analytics.

Visual Clutter and Design Trade-offs. In responsive design, visualization transformations adjust visual encoding from points and lines to areas to better represent data at far distances. However, we acknowledge that this approach can introduce challenges like area overlap, which may obscure key information and hinder accurate data interpretation. A typical example of this issue arises when transforming a line chart into a bold line chart. In the original line chart, individual data points and trends are distinguishable, but after the visualization transformation without data transformation, overlapping areas can create visual clutter. This is especially problematic when multiple datasets are represented. To mitigate this, future designs might explore techniques like layering, transparency, or alternative chart types. Nonetheless, these solutions involve trade-offs in complexity and interpretability, highlighting the need for careful consideration when applying visualization transformations.

Data transformation with visual change is attracting. This work highlights coherence as vital for responsive SitV design. While scatter plot sampling effectively reduced mental load (DataOnly: $M = 3.65$, VisualOnly: $M = 5.38$, Combined: $M = 6.03$ vs. Static: $M = 2.81$), participants showed low awareness of these changes in the data-only condition, leading to less pronounced subjective gains. In contrast, sampling in line charts and aggregation in bar charts produced more noticeable visual changes and better performance (e.g., Line Chart Combined: $M = 5.73$

vs. Static: $M = 2.88$; DataOnly: $M = 4.58$; VisualOnly: 4.31). However, as seen with line charts, noticeable changes did not always enhance efficiency. Balancing coherence and mental load is key, ensuring transformations are consistent and discernible.

Situatedness is preserved in transformations. To ensure the preservation of situatedness, our design maintains consistent spatial and semantic connections between the visualization and its referent, specifically by avoiding unintentional alterations to the anchor position or orientation during transformations. The results from both User Study 1 and the newly added User Study 2 (Dual Referents) indicate that participants found the responsive designs effective in situated contexts. Even in the dual-referent scenario, where participants had to distinguish between two physical objects, they were able to easily locate the target referent and interpret the associated data without disorientation. This confirms that our responsive mechanism preserves the essential spatial link required for situated analytics. Future work could further explore how these spatial layouts adapt to more complex social contexts, such as *community* or *collaborative activity* (Bressa et al. 2022; Martins et al. 2023).

Limitations and Future works. The study’s findings are constrained by several technical and design limitations, which may affect the generalizability of the results. First, the resolution limitations of the HoloLens 2 can adversely impact the legibility of visual elements, especially at farther distances. User feedback on scatter plot sampling indicates that the visibility and legibility of scatter points decreased with distance, compromising the transformation’s impact. This may limit the applicability of our findings to devices with higher display fidelity or different display form factors. However, the responsive SitV design remains usable due to its fundamental logic. Second, the transformations were evaluated using only chart-based visualizations with a single referent. While our results demonstrate promising effects in controlled settings, their applicability to more complex, dynamic, or multi-referent environments remains uncertain. For example, in urban-scale AR applications with multiple overlapping SitVs, spatial disambiguation and layout adaptation may require additional mechanisms not tested in our study. Third, we focused primarily on visual element transformations, without transforming textual components. Given that many real-world SitV applications involve textual cues (e.g., annotations, labels, legends), the lack of text-aware transformation may reduce ecological validity in these contexts. Taken together, these limitations suggest that while the core mechanisms of responsive transformation show general promise, their effectiveness may vary across hardware, data types, and AR deployment contexts.

Looking ahead, we aim to enhance specific transformations within our design space for future work. Based on user

feedback, we will increase the size of scatter points post-sampling or aggregate them into bubbles to better represent distribution and density. We also intend to explore the balance between sampling degree and overstriking threshold for line charts to maximize line thickness while retaining the ability to reflect main fluctuations. Additionally, we plan to extend our design space to include transformations for textual elements, recognizing the importance of text in information retrieval during chart reading. We will consider text transformation in terms of size, layout, and density to optimize readability and user comprehension.

7 Conclusion

This study introduces a responsive design mechanism aimed at achieving effective situated visualization based on viewing distance. The study addresses the existing gap in research regarding the necessity for responsive SitV design across diverse scenarios such as equipment management and book borrowing, whilst responsive design tailored to SitV has been lacking. The design mechanism has been developed based on user feedback collected during a preliminary study. Users provided feedback that was categorized into design scopes of visualization, interaction, and layout, along with considerations regarding legibility, readability, situatedness, and coherence. A prototype SitV design has been implemented, utilizing data and visualization transformations across four common chart types. To validate the effectiveness of our responsive SitV design in terms of information understanding and user engagement, two user studies involving 14 and 12 participants were conducted. Quantitative results indicate that the responsive SitV design enables users to comprehend information more rapidly and accurately. These findings suggest practical implications for the development of SitV applications, particularly in contexts where users interact with data across varying distances or physical environments. Looking ahead, the responsive SitV design mechanism could be further refined through adaptive personalization techniques or extended to support more complex visualization types and multi-user interactions in collaborative settings.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10055-026-01370-x>.

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Author contributions Z.Y. wrote the main manuscript, developed the systems, and mainly conducted the user study. Z.W., Y.H., and Q.Z. helped develop the system and conducted the user study. W.Z. initiated

the idea and offered support and guidance for this work. All authors reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

Ethical approval and consent to participate the preliminary study and the user study. Informed consent was obtained from all individual participants included in the study.

Consent for publication All participants consented to the publication of their anonymized data as part of this research.

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